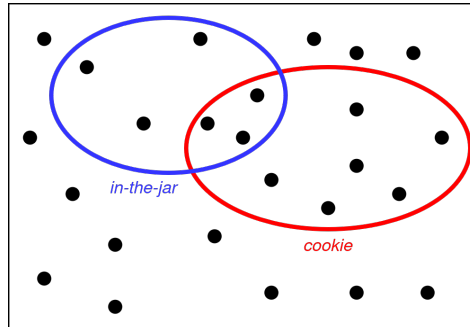


Semantic Theory 2026: Exercise 4 Key

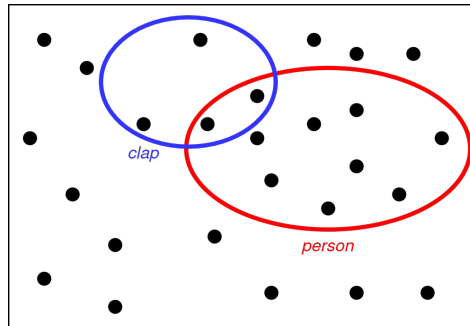
Question 1

Give the truth conditions for the following sentences using Generalized Quantifier Theory: interpret each VP as a property (i.e. a set of entities) and interpret quantified noun phrases (underlined) as sets of properties. Illustrate your answer with a graphical depiction of a model in which the sentence is true.

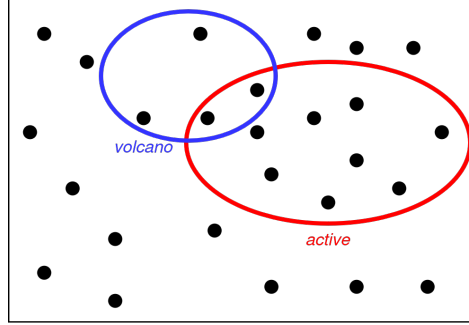
- a. $\llbracket \text{at least three cookies are in the jar} \rrbracket = 1$ iff
 $\llbracket \text{in-the-jar} \rrbracket \in \{P \subseteq U_M \mid |\llbracket \text{cookie} \rrbracket \cap P| \geq 3\}$ iff
 $|\llbracket \text{cookie} \rrbracket \cap \llbracket \text{in-the-jar} \rrbracket| \geq 3$



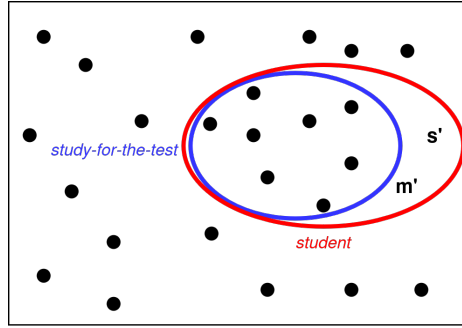
- b. $\llbracket \text{few people clapped} \rrbracket = 1$ iff
 $\llbracket \text{clap} \rrbracket \in \{P \subseteq U_M \mid |\llbracket \text{person} \rrbracket \cap P| \leq \frac{|\llbracket \text{person} \rrbracket|}{n}\}$ iff
 $|\llbracket \text{person} \rrbracket \cap \llbracket \text{clap} \rrbracket| \leq \frac{|\llbracket \text{person} \rrbracket|}{n}$



- c. $\llbracket \text{some, but not all, volcanoes are active} \rrbracket = 1$ iff
 $\llbracket \text{active} \rrbracket \in \{P \subseteq U_M \mid 1 \leq |\llbracket \text{volcano} \rrbracket \cap P| < |\llbracket \text{volcano} \rrbracket|\}$ iff
 $1 \leq |\llbracket \text{volcano} \rrbracket \cap \llbracket \text{active} \rrbracket| < |\llbracket \text{volcano} \rrbracket|$



- d. $\llbracket \text{every student except for Mark and Sally studied for the test} \rrbracket = 1$ iff
 $\llbracket \text{study-for-the-test} \rrbracket \in \{P \subseteq U_M \mid \llbracket \text{student} \rrbracket - \{\llbracket m' \rrbracket, \llbracket s' \rrbracket\} \subseteq P\}$ iff
 $\llbracket \text{student} \rrbracket - \{\llbracket m' \rrbracket, \llbracket s' \rrbracket\} \subseteq \llbracket \text{study-for-the-test} \rrbracket$



Question 2

Determine the monotonicity properties (left and right) of the following determiners. Show how you derived these monotonicity properties.

- a. *half of the*

Right upward:

half of the students walked quickly $\models$ *half of the students walked*

Not left monotonic:

half of the students walked $\not\models$ *half of the people walked*

half of the people walked $\not\models$ *half of the students walked*

- b. *no more than six*

Right downward:

no more than six students walked $\models$ *no more than six students walked quickly*

Left downward:

no more than six people walked $\models$ *no more than six students walked*

c. *most*

Right upward:

most people swam quickly $\models$ *most people swam*

Not left monotonic:

most people swam $\not\models$ *most students swam*

most students swam $\not\models$ *most people swam*

Question 3

Prove that the following statement holds:

The external negation of a right downward monotonic (monotone decreasing) quantifier is a right upward monotonic (monotone increasing) quantifier

$DM(Q) := \forall P, R: (\llbracket P \rrbracket \subseteq \llbracket R \rrbracket \wedge Q(R)) \rightarrow Q(P)$

$UM(Q) := \forall P, R: (\llbracket P \rrbracket \subseteq \llbracket R \rrbracket \wedge Q(P)) \rightarrow Q(R)$

(i) assume we have P, R such that $\llbracket P \rrbracket \subseteq \llbracket R \rrbracket$ and $\neg Q(P)$

(ii) assume $\neg Q(R)$ does not hold, i.e. $\neg(\neg Q(R))$, i.e. $Q(R)$

(iii) then by the assumption that $\llbracket P \rrbracket \subseteq \llbracket R \rrbracket$ and $DM(Q)$, we have $Q(P)$

(iv) by (i) and (iii), we have $Q(P) \wedge \neg Q(P)$. This is a contradiction.

Therefore, $DM(Q) \rightarrow UM(\lambda P. \neg Q(P))$